UNIVERSITATEA TEHNICĂ DIN CLUJ-NAPOCA
Facultatea de Electronică, Telecomunicații și Tehnologia Informației
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COMPUTER AIDED DESIGN PROJECT

VOLTAGE REGULATOR WITH OVERCURRENT PROTECTION

Professors:
Decan prof. dr. ing. Pop Ovidiu Aurel
S.l. dr. ing. Ștețco Elena Mirela

Student: Silaghi Sergiu
Year: II
Group: 2123

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1. Project requirements

Design a voltage regulator where the output voltage is adjustable in the range $V_{out_min} = 6V$ and $V_{out_max} = 19V$, and the output current is limited to the maximum value $I_{o_lim} = 1A$, knowing that the input voltage is $V_{in} = 24V$.

2. Technical details

- Output voltage in the range [6V ; 19V]
- Output current limited to 1A
- Output load resistance varies in the range [0Ω ; 100kΩ]
- Input voltage is 24V
- The reference voltage is set using a 1N4733 Zener diode with a value of 5.1V
- Error amplifier: TL082
- Regulation transistor: Q2N2222
- Protection transistor: Q2N3055
- Resistors: from E96 series

3. Block diagram

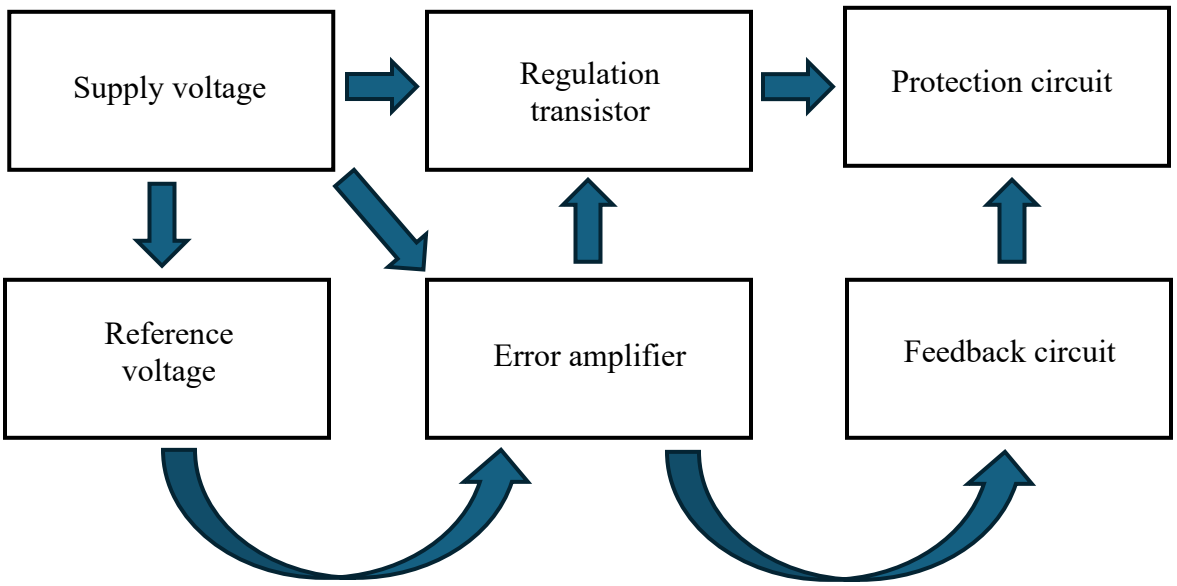

Figure 1. Block diagram of the circuit

4. Electrical schematic

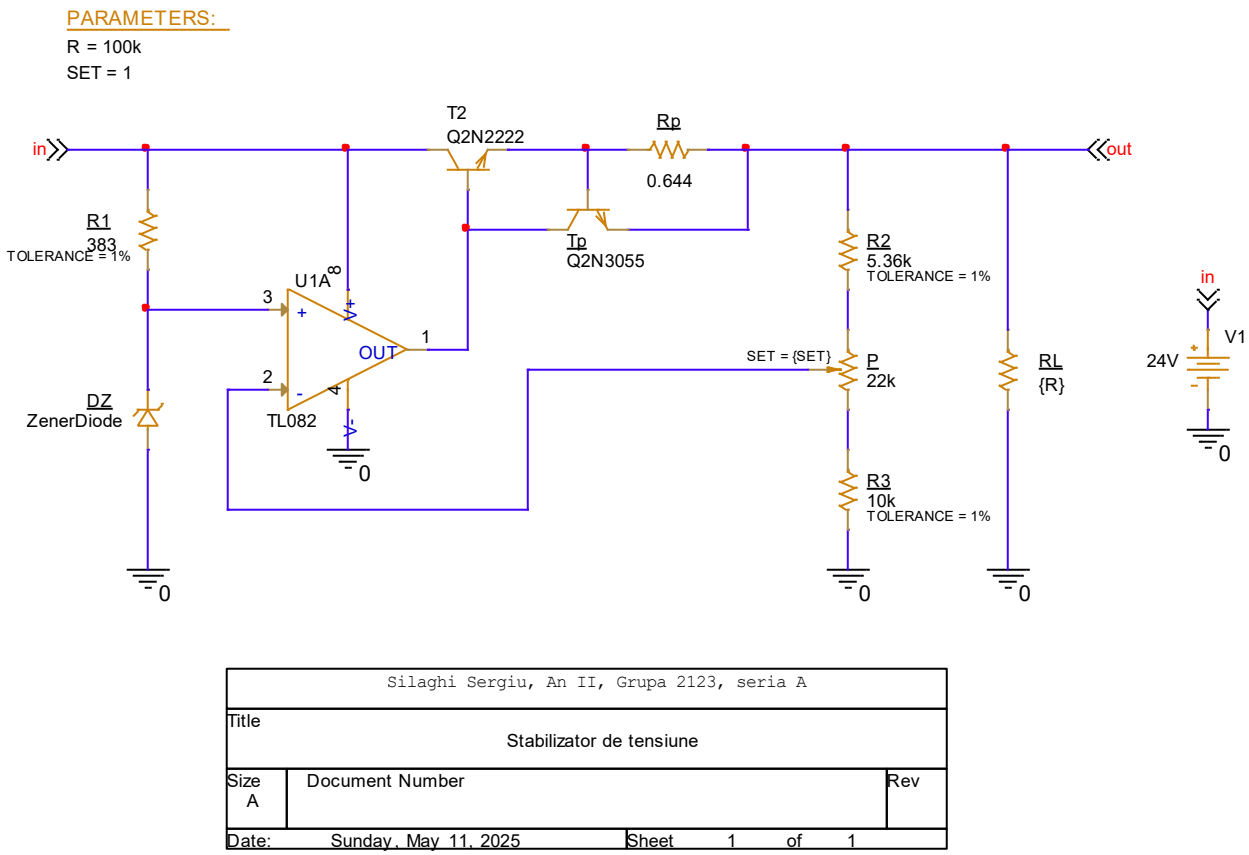

Figure 2. Electrical schematic of the circuit

5. Detailed circuit design and calculation

5.1. Supply voltage + reference voltage

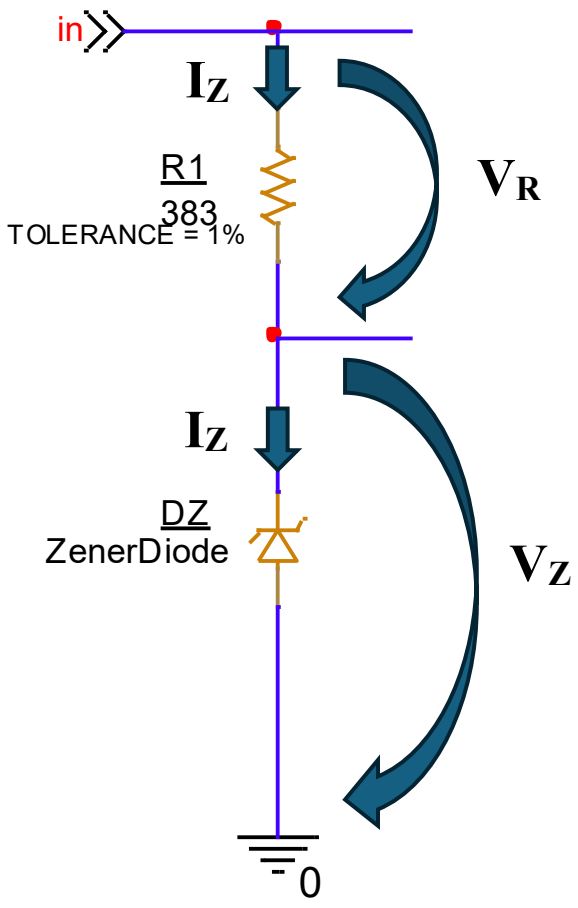

Figure 3. Supply voltage + reference voltage

The supply voltage in figure 3 represents the voltage applied to the circuit input, and the reference voltage achieved with a Zener diode, also in Figure 3, is an essential value with which we can stabilize the output voltage, the resistor R_1 having the role of protecting the diode.

5.1.1. Sizing of R_1 resistor

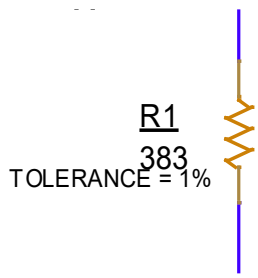

Figure 4. R_1 resistor

Looking at the datasheet of the 1N4733 Zener diode, it has a stabilization voltage V_z=5.1V and a Zener current I_z=49mA. Knowing these aspects, we can calculate the value of the resistor R_1 in figure 4:

$$R_1 = \frac{V_{in} - V_z}{I_z} = \frac{24V - 5.1V}{49mA} = 385.71\Omega \qquad (1)$$

We do not have a standard resistance with the calculated value, but a close value is found in the E96 series, namely 383Ω. In this case, the current through the Zener diode, calculated using equation (1), becomes:

$$I_z = \frac{V_{in} - V_z}{R_1} = \frac{24V - 5.1V}{383\Omega} = 49.3mA$$

With these values of the current through the Zener diode and the resistance, substituting in equation (1) we obtain a voltage V_z equal to 5.12V. This value will have an influence on the stabilization of the output voltage but to a small extent.

To avoid any inconvenience, we can make the following configuration for resistor R_1 using standard resistor values from the E96 series:

$$374\Omega + 11.8\Omega \Rightarrow R_{ech} = 385.8\Omega$$

With this configuration we get very close to the mathematically calculated value for R_1 and we obtain some negligible differences from the catalog sheet, both for V_z and I_z.

5.1.2. Zener diode modeling

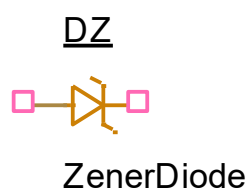

Figure 5. Zener diode modeling

To model the Zener diode in figure 5 and represent its characteristics, we took the actual data from the 1N4733 diode datasheet. All information about the diode is given at an ambient temperature of 25°C in the datasheet.

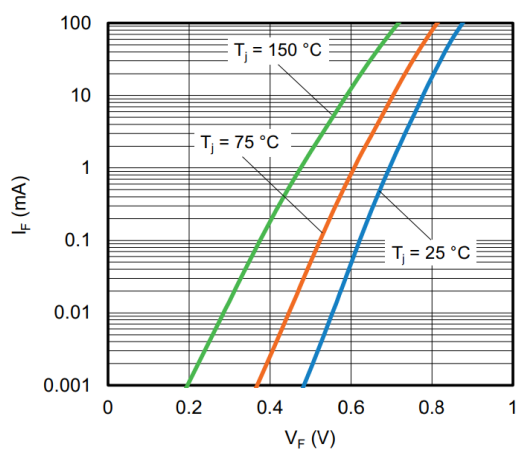

Figure 6. I-V characteristic of the diode at different ambient temperatures

OrCAD performs simulations at an ambient temperature of 27°C, but, as can be seen from the I-V characteristic in figure 6, there is a small difference between the representation at 25°C and 75°C, which means that the differences between 27°C in OrCAD and 25°C in figure 6 will be negligible.

From the I-V characteristic in figure 6, it appears that at voltages lower than 0.7V we have very small currents, thus, we can say that the diode starts conducting from 0.7V. Also, in forward bias, it is specified in the datasheet that at a current of 200mA we can have a maximum of 1.2V, which must be found on the characteristic.

In the reverse bias characteristic, we are interested in Reverse Leakage and Reverse Breakdown in order to represent it in OrCAD. For Reverse Leakage we find out from the catalog sheet that at a voltage of 1V we must have a current less than 10uA, so at voltages lower than the Zener voltage we must have currents almost 0, and for Reverse Breakdown, we find out that when the diode starts to stabilize, the current must increase so that at a value of 5.1V we must have a Zener current of 49mA.

At the same time, in OrCAD, for Reverse Breakdown we must fill in the value of an impedance Z_z, which we know from the datasheet must be < 7Ω. To calculate this impedance we use the model of a 1N4733 diode default in OrCAD, for which an impedance Z_z= 2.2Ω is given at a current of 20mA. We calculate $Z_z$in our case:

$$\left.\begin{array}{l} 2.2\Omega \ldots .20mA \\ Z_z \ldots \ldots .49mA \end{array}\right\} \quad Z_z = \frac{49mA \cdot 2.2\Omega}{20mA} = 5.39\Omega$$

After completing these steps, the diode is fully modeled and we will be able to represent both the forward and reverse bias characteristics.

5.2. Error amplifier + feedback circuit

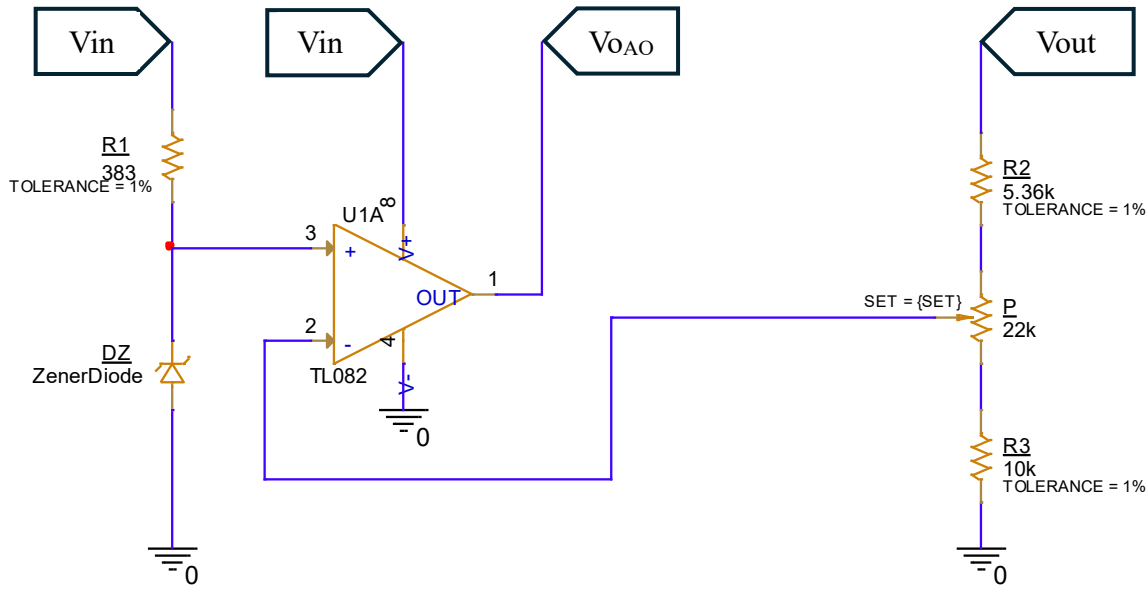

Figure 7. Error amplifier + feedback circuit

Vo_{AO} – output voltage of the Op-Amp

We chose the TL082 operational amplifier in figure 7 because its datasheet states that it can handle a supply voltage of up to 36V without any problems. This is an error amplifier and is a very important element because it connects the reference voltage connected to the non-inverting terminal and the feedback circuit connected to its inverting terminal. The operational amplifier has negative feedback, which means that the relationship v+ = v- (2) occurs.

$$\begin{cases} v^+ = V_z \\ v^- = \dfrac{\alpha P + R_3}{R_2 + R_3 + P} \cdot V_{out} \end{cases}$$

According to (2 $\Rightarrow$ $\frac{\alpha P + R_3}{R_2 + R_3 + P} \cdot V_{out} = V_z$ $\Rightarrow$ $V_{out} = \frac{R_2 + R_3 + P}{aP + R_3} \cdot V_z$ (3)

In the feedback circuit (figure 7) there is a variable resistor (P), thanks to which we can adjust the output voltage within a certain range of values, depending on the position of the cursor α. From equation (3), we note that the output voltage depends on both the reference voltage and the value of α.

Expression of V_{out} depending on α, where α ∈ [0, 1], according to (3), is:

$$\alpha = 1 \Rightarrow V_{out_min} = \frac{R_2 + R_3 + P}{P + R_3} \cdot V_z \qquad (4)$$

$$\alpha = 0 \Rightarrow V_{out_max} = \frac{R_2 + R_3 + P}{R_3} \cdot V_z \qquad (5)$$

We need to stabilize the output voltage in the range [6V; 19V], so we have V_{out_min}=6V and V_{out_max}=19V, thus we will obtain a system with two equations with which we can size the components so that the circuit stabilizes the output voltage in the desired range.

For V_{out_min}, according to (4) we have:

$$6V = \frac{R_2 + R_3 + P}{P + R_3} \cdot 5.1V \Rightarrow \frac{R_2 + R_3 + P}{P + R_3} = 1.17 \qquad (6)$$

For V_{out_max}, according to (5), we have:

$$19V = \frac{R_2 + R_3 + P}{R_3} \cdot 5.1V \Rightarrow \frac{R_2 + R_3 + P}{R_3} = 3.72 \qquad (7)$$

The system formed by equations (6) and (7) is:

$$\begin{cases} \dfrac{R_2 + R_3 + P}{P + R_3} = 1.17 \\ \\ \dfrac{R_2 + R_3 + P}{R_3} = 3.72 \end{cases}$$

From this system we will extract the equation $R_2 + R_3 + P$ in terms of R_3 from equation (7) and then we will substitute into equation (6) to extract R_3in terms of P:

From (7) = $R_2 + R_3 + P = 3.72 \cdot R_3$ (8)

From (6) ⇒ $\dfrac{3.72 \cdot R_3}{P + R_3} = 1.17 \Rightarrow 2.55 \cdot R_3 = 1.17 \cdot P \Rightarrow R_3 = 0.459 \cdot P$ (9)

R_2 in terms of P:

According to (8) and (9) ⇒ $R_2 + 0.459 \cdot P + P = 1.707 \cdot P \Rightarrow R_2 = 0.248 \cdot P$ (10)

We size the components using (9) and (10):

Fie $P = 22k\Omega$ => $\begin{cases} R_3 = 0.459 \cdot 22k\Omega \Rightarrow R_3 = 10.09k\Omega \\ R_2 = 0.248 \cdot 22k\Omega \Rightarrow R_2 = 5.45k\Omega \end{cases}$

The 22kΩ potentiometer P is a standard value, but the calculated values of the resistors R_3 and R are not standard, so we will replace the resistor R_3 with a 10kΩ one, and R_2 with a 5.36kΩ one, both resistor values being standard values from the E96 series.

5.3. Regulation transistor + protection circuit

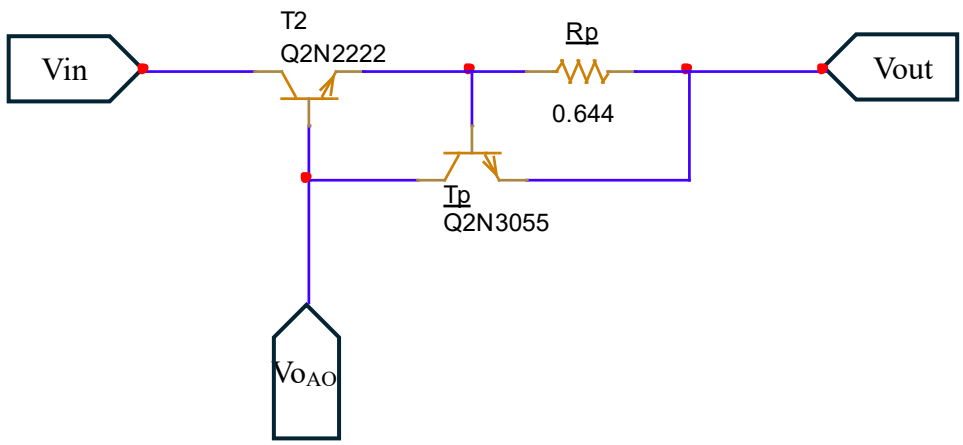

Figure 8. Regulation transistor + protection circuit

R_p sizing:

$$R_p = \frac{V_{BE_Tp}}{I_{o_lim}} = \frac{0{,}65V}{1A} = 0{,}65\Omega \qquad (11)$$

With the calculated value of the resistance R_p from equation (11), we obtain an output current of 1A only in the ideal case, when V_{BE_Tp}= 0.65V. In fact, this voltage is lower, approximately 0.644V, so we obtain a protection resistance R_p=0.644Ω in order to limit the output current to I_{o_lim}=1A. This resistance is not a standard one, but the closest standard value is 0.649Ω with a tolerance of ±1%.

Transistor T2 Q2N2222 (figure 8) is an NPN type control transistor and its role is to amplify the current coming from the output of the error amplifier. The current amplified by transistor T2 is monitored by resistor R_p (figure 8), which is a protection resistor with a fixed value depending on the limit current that we want to obtain. As long as the current passing through resistor R_p does not exceed the limit current, transistor Tp (figure 8) remains OFF.

When the current through the circuit increases above the current limit value, the transistor Tp Q2N3055 enters conduction, which is a protection and power transistor, NPN type, chosen so that, in our case, it can carry a current of 1A. From the datasheet of this transistor we find that it can carry up to 15A, so it is a good choice. When the protection comes into operation, the output current is limited to the current value I_{o_lim}, and the output voltage begins to decrease, protecting the circuit.

5.4. Load resistance

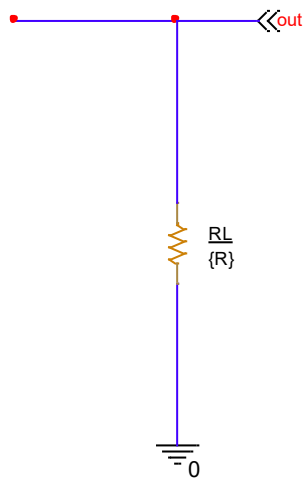

Figure 9. Load resistance R_L

Using the load resistor R_L in figure 9, we can correctly visualize the decrease in output voltage with current limitation, depending on different values of R_L. Due to this fact, the resistance R_L does not have a fixed value, but varies according to a parameter {R} that takes values in the range [0Ω; 100kΩ].

6. Simulations

6.1. I-V characteristic in forward bias of the Zener diode

To see if the diode is modeled correctly in forward bias, the characteristic in figure 12 must comply with the specifications in the datasheet mentioned in point 5.1.2., and for this we created the following circuit (figure 10) in which we connected only the diode.

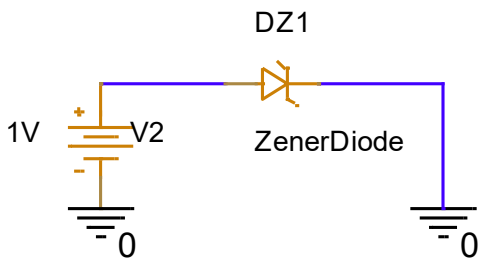

Figure 10. The circuit consisting only of the diode

The characteristic in figure 12 was obtained from the simulation tab through a main DC Sweep analysis where we swept the input source from 0V to 2V with a step of 0.1 (figure 11), then we plotted the current through the diode (figure 12).

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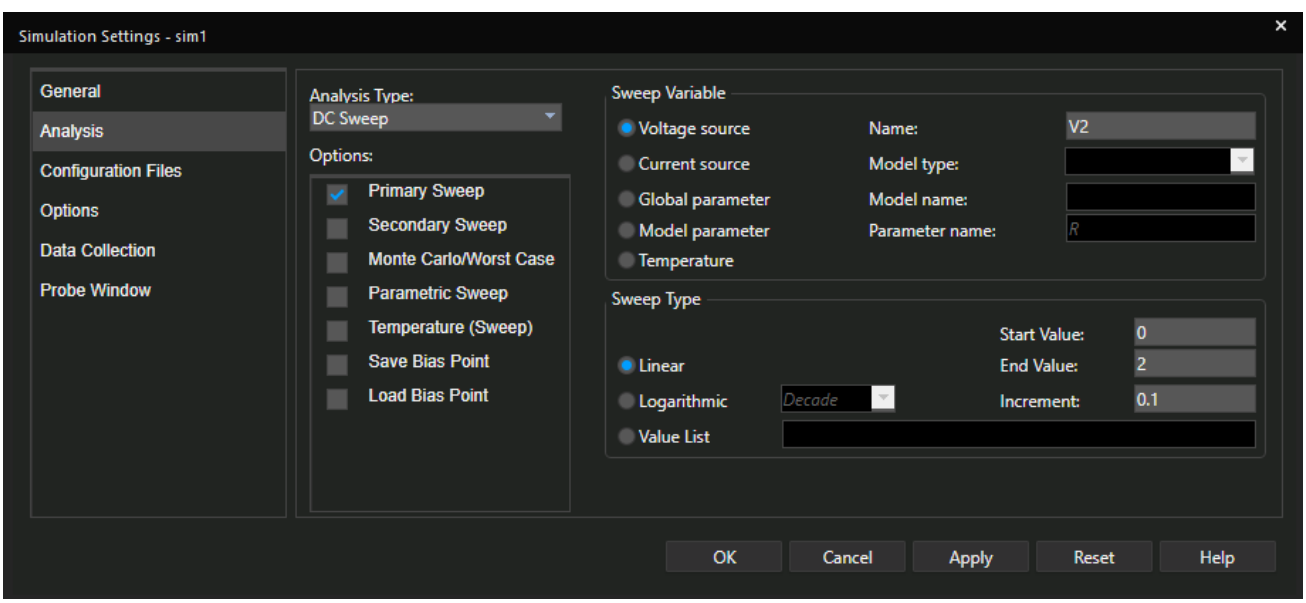


Figure 11. DC Sweep analysis

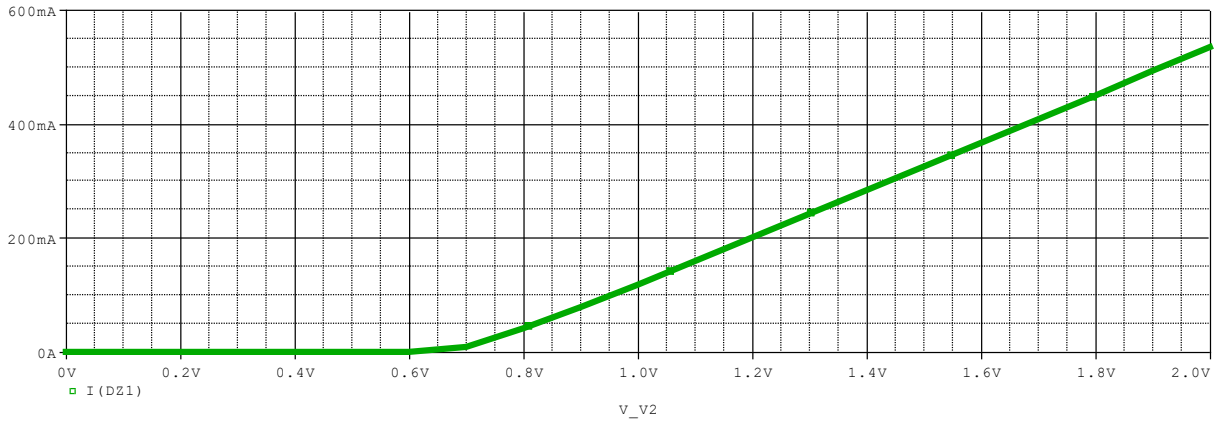


Figure 12. I-V characteristic in forward bias of the Zener diode

As we can see from the graph, the forward-biased diode works correctly because it has a threshold voltage of 0.7V, and at a current of 200mA we have a voltage of 1.2V, which complies with the data sheet.

6.2. I-V characteristic in reverse bias of the Zener diode

To see if the diode is correctly modeled in reverse bias, the characteristic in figure 14 must comply with the scpecifications from the datasheet mentioned in point 5.1.2. .

The representation of the characteristic in figure 14 was generated from the simulation tab by sweeping the input source between 0V and 24V with a step of 0.1 based on a DC Sweep analysis (figure 13), then we plotted the voltage across the diode (figure 14 in green) and the current through the diode (figure 14 in red).

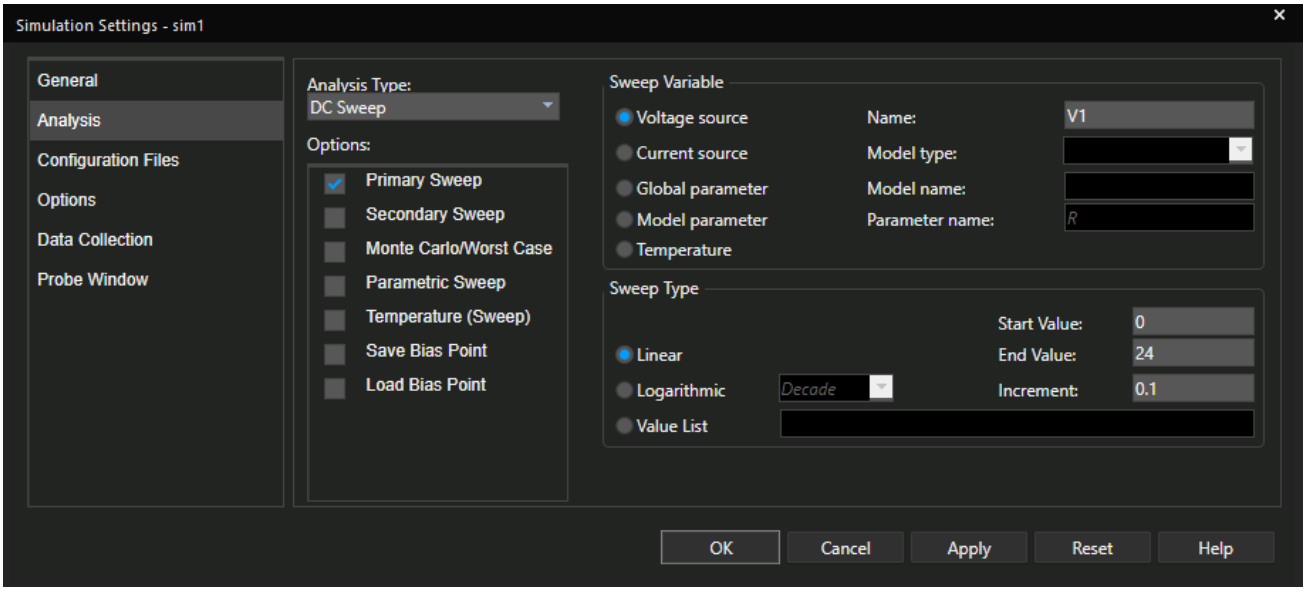


Figure 13. DC Sweep analysis

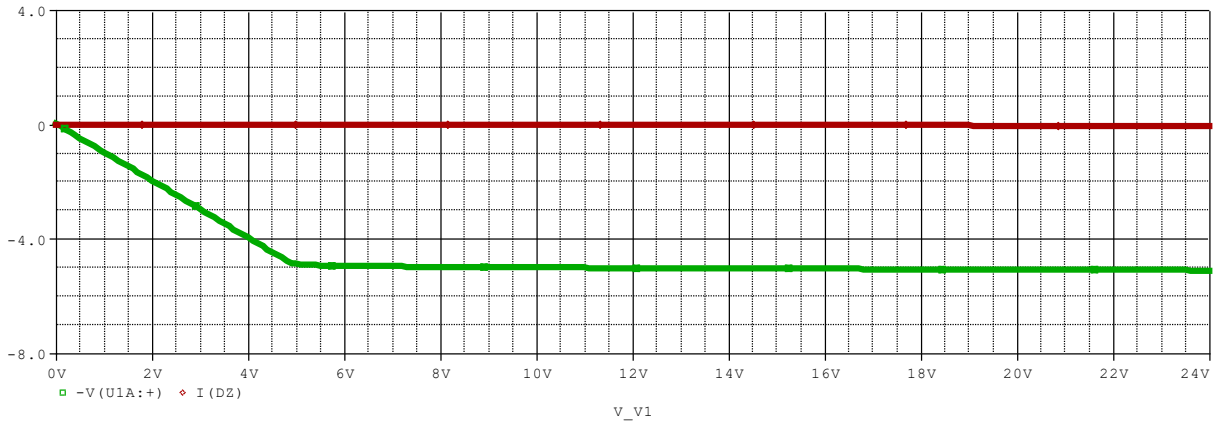


Figure 14. I-V characteristic in reverse bias of the Zener diode

Probe Cursor

Trace Color	Trace Name	Y1	Y2	Y1 - Y2		Y1(Cursor1) - Y2(Cursor2)		-5.1196		
	X Values	23.867	0.000	23.867		Y1 - Y1(Cursor1)	Y2 - Y2(Cursor2)	Max Y	Min Y	Avg Y
CURSOR 1,2	-V(U1A:+)	-5.1000	19.614m	-5.1196		0.000	0.000	19.614m	-5.1000	-2.5402
	I(DZ)	-49.014m	62.805n	-49.014m		5.0510	-19.614m	62.805n	-49.014m	-24.507m

Figure 15. Value of current I_z at stabilization voltage V_z

As we can see from figure 15, the reverse biased diode works correctly because for voltages lower than the Zener voltage we have currents almost 0, and once we reach the Zener voltage the current increases so that at 5.1V we have 49mA (figure 15). The current increase is not very visible in figure 14, but from figure 15 the previously mentioned values emerge very clearly.

6.3. Limiting the output voltage to V_{out_min}

To represent the characteristic In figure 17, we set a main Transient analysis with a simulation duration of 50ms (figure 16), with the potentiometer value set to 1.

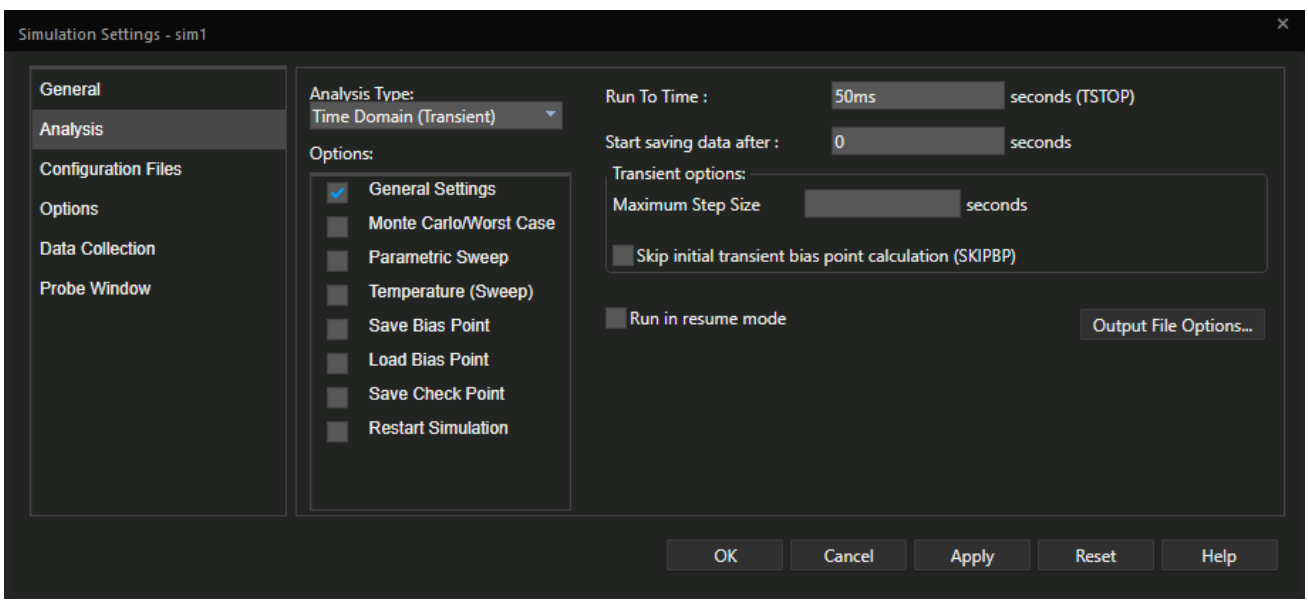


Figure 16. Transient analysis

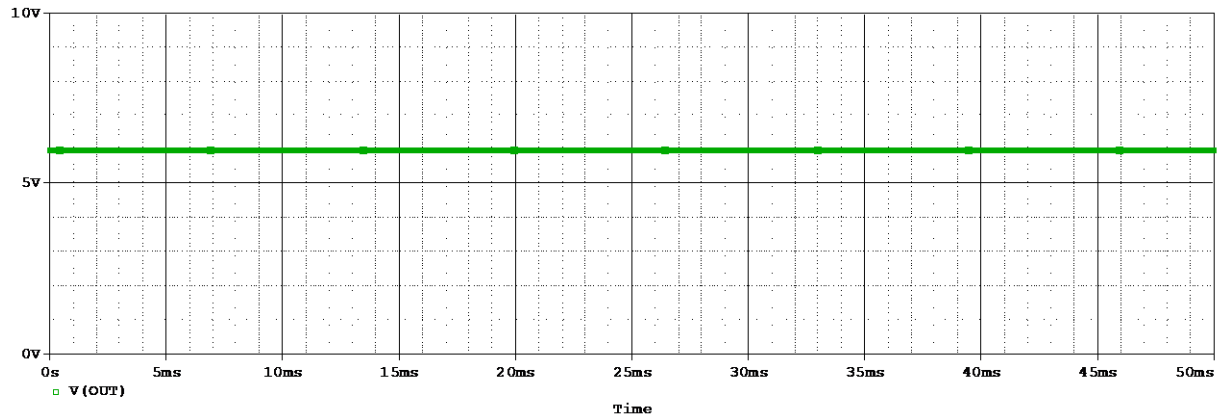


Figura 17. Output voltage limited to V_{out_min}

6.4. Limiting the output voltage to V_{out_max}

To represent the characteristic In figure 19, we set a main Transient analysis with a simulation duration of 50ms (figure 18), with the potentiometer value set to 0.

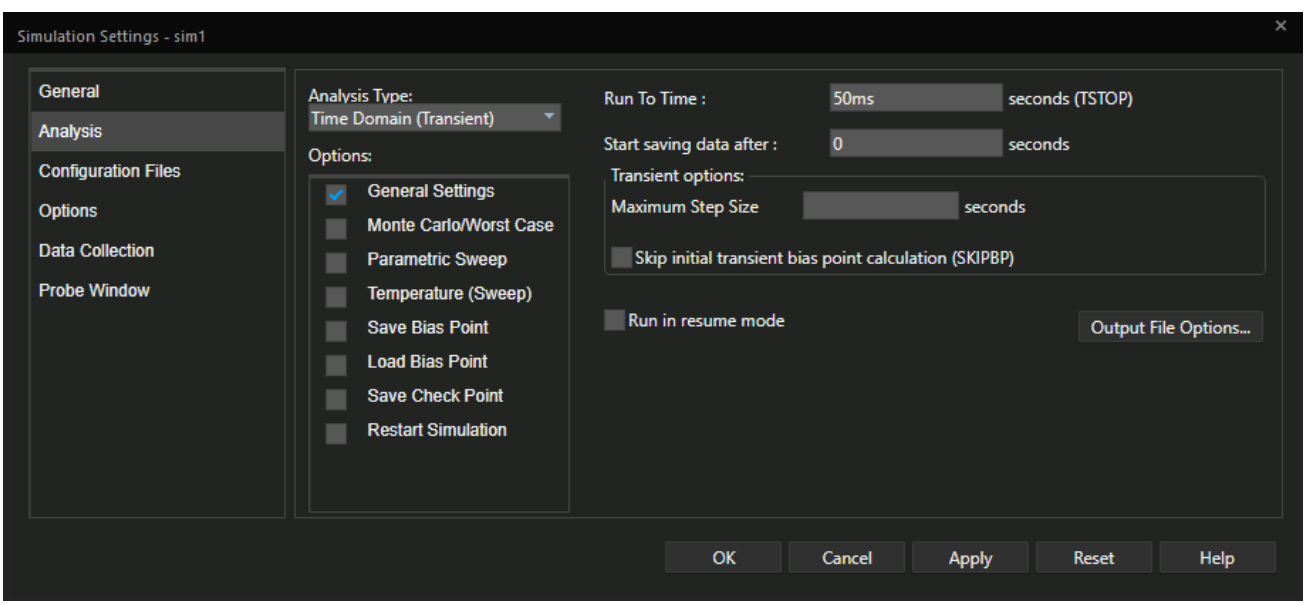


Figure 18. Transient analysis

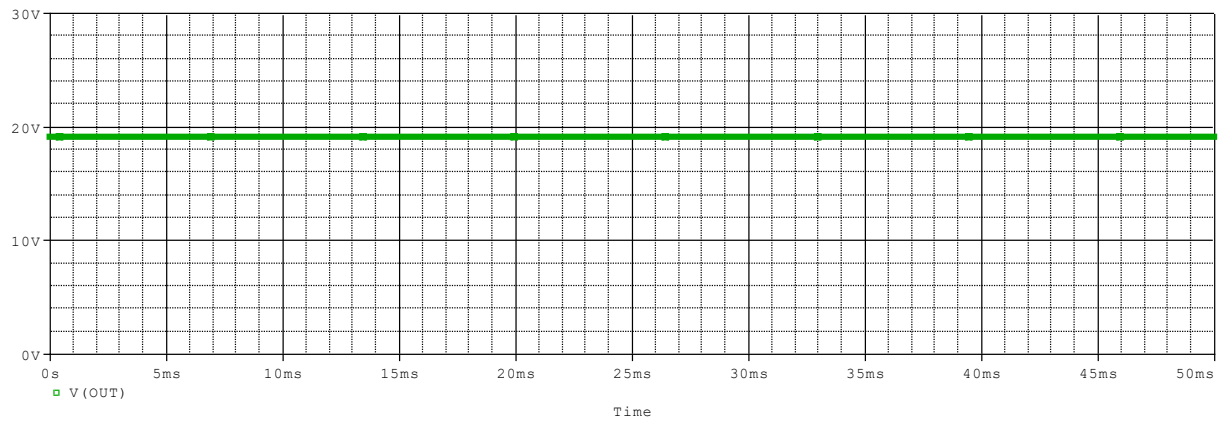


Figure 19. Output voltage limited to V_{out_max}

6.5. Output voltage variation depending on the potentiometer cursor value

For this step, we set a main Transient analysis with a simulation duration of 50ms (figure 20) along with a secondary Parametric Sweep analysis where we swept the potentiometer`s SET parameter from 0 to 1 with a step of 0.1 (figure 21).

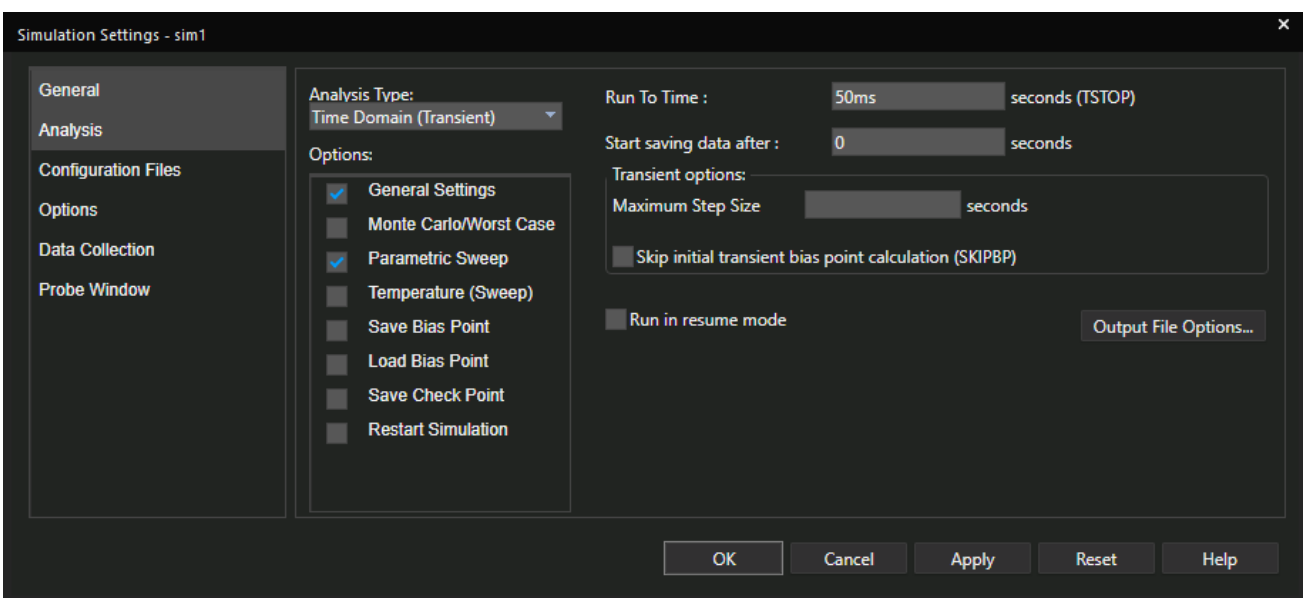


Figure 20. Transient analysis

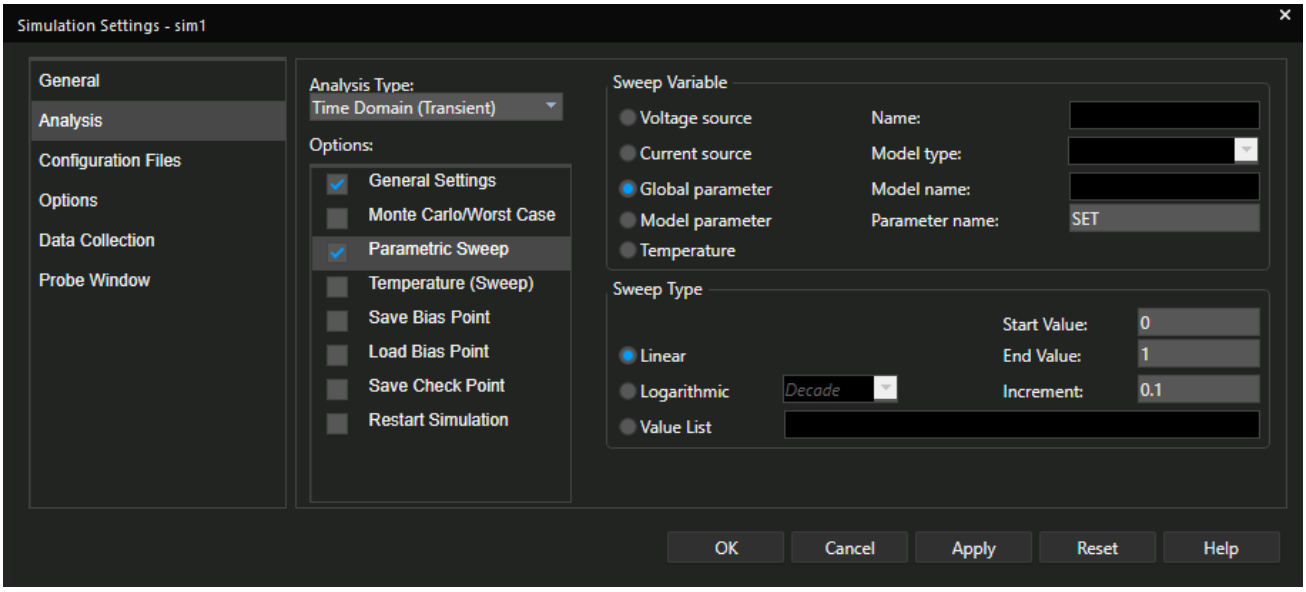


Figure 21. Parametric Sweep analysis

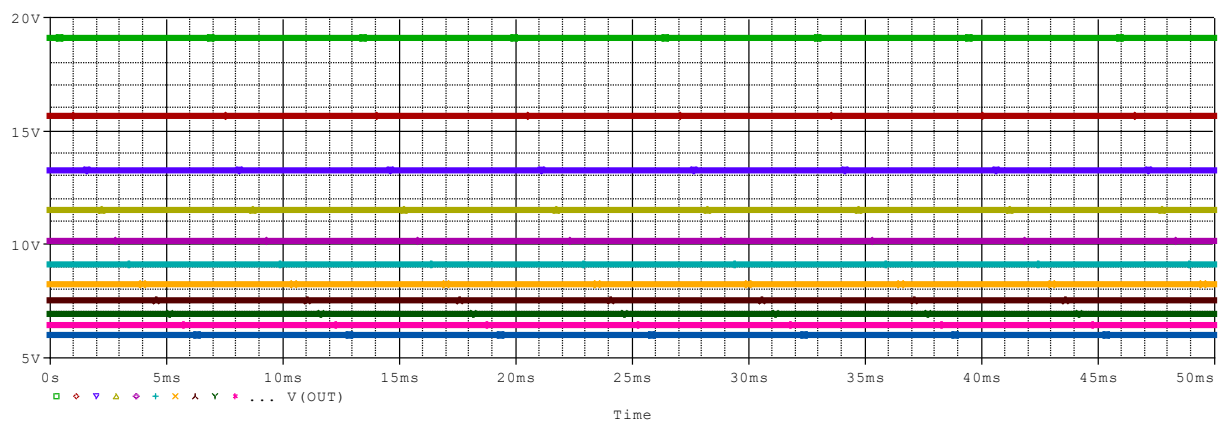


Figure 22. Output voltage variation depending on the potentiometer`s cursor position

To see the simulation results depending on the potentiometer cursor, i.e. the cursor being on the Ox axis, we ran a performance analysis (figure 23).

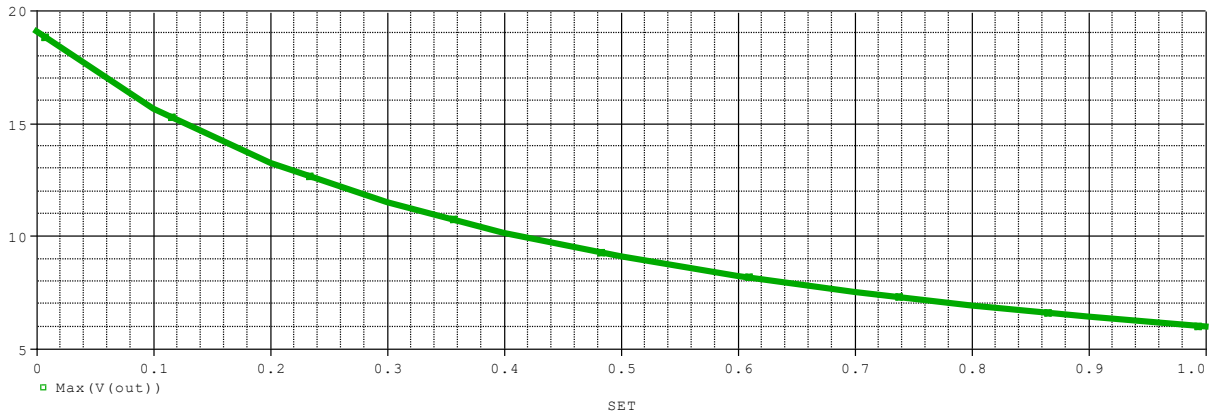


Figure 23. Output voltage variation depending on the potentiometer`s cursor position

The characteristic generated in figure 23 is correct because it indicates different values of the output voltage depending on the variation of the SET cursor of the potentiometer so that when the cursor is at 0 we have a maximum voltage of 19V, and when the cursor is at 1 we have a minimum voltage of 6V. We notice that in figure 22, 11 different values were generated for V_{out}, and this happens because in the interval [0; 1] we have 11 values for a step of 0.1.

6.6. Current-voltage limiting characteristic

To see if the protection circuit is working properly , we generated the characteristic in figure 26 by setting a main Transient analysis with a simulation duration of 50ms (figure 24) and a secondary Parametric Sweep analysis in which we varied the load resistance between 0Ω and 100kΩ with a step of 100 (figure 25), then we plotted the output (figure 26 in green) and the current through the load resistance (figure 26in red). We chose a larger range for the resistance to have a better view of the characteristic.

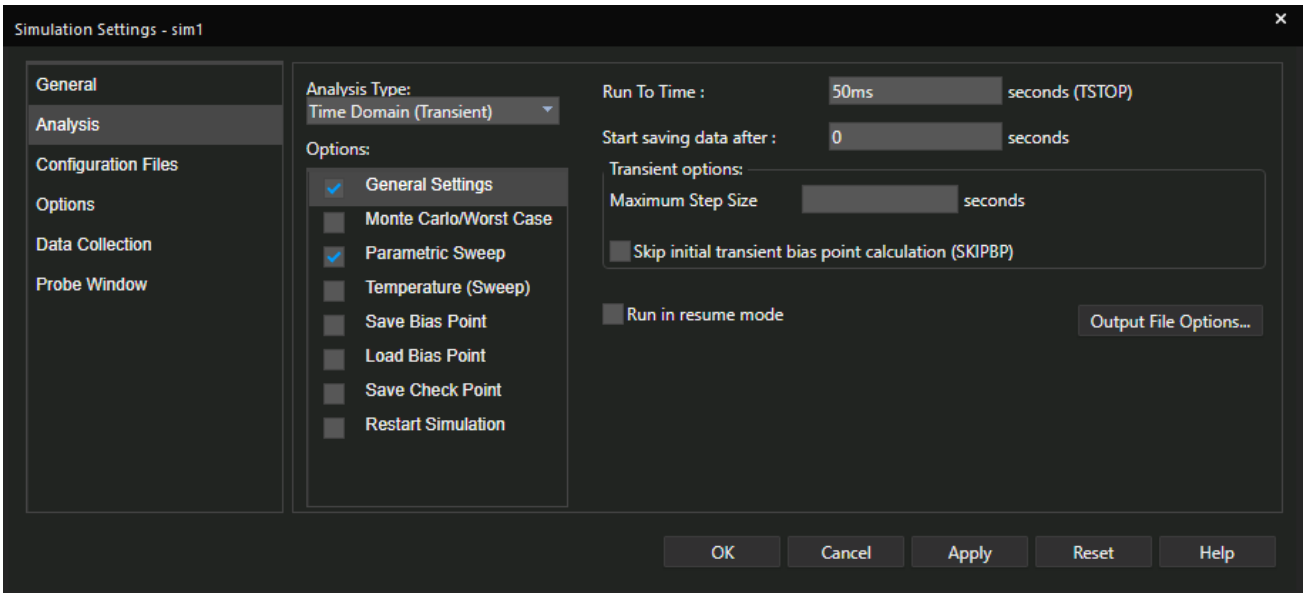


Figure 24. Transient analysis

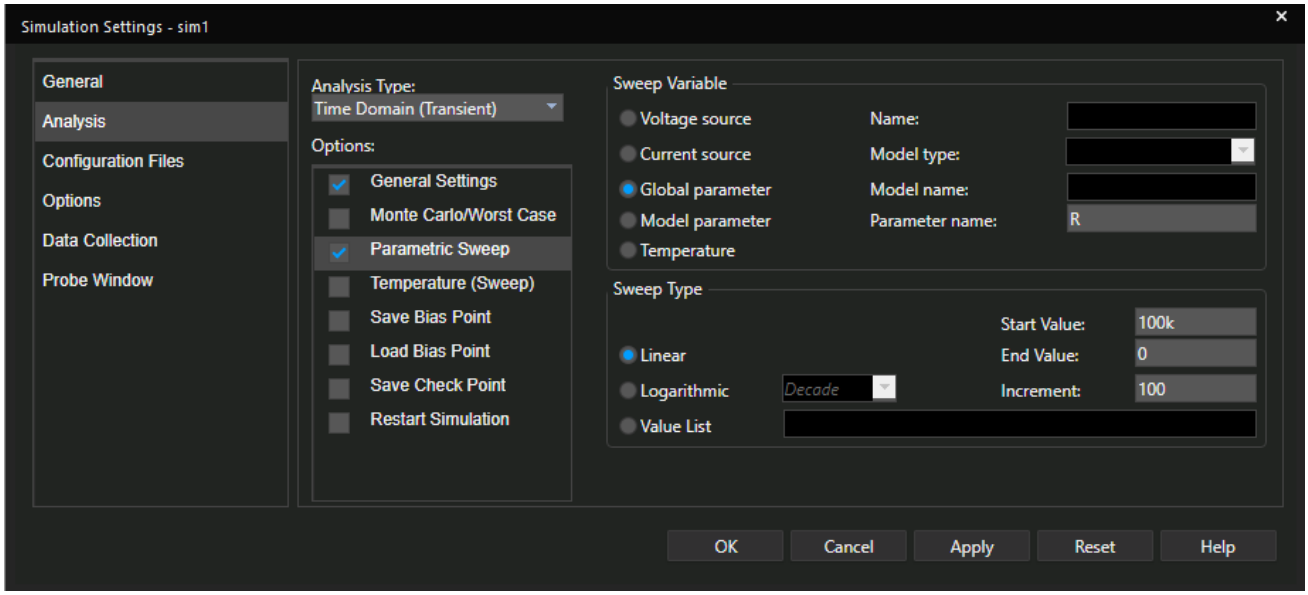


Figure 25. Parametric Sweep analysis

Through a performance analysis, we can see the characteristic in figure 26.

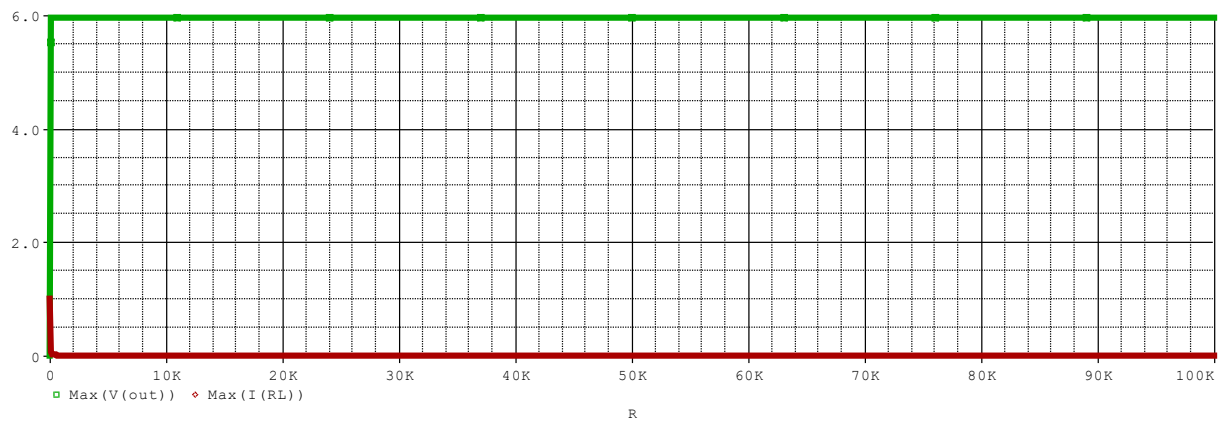


Figure 26. Current-voltage limiting characteristic

Probe Cursor

	Trace Color	Trace Name	Y1	Y2	Y1 - Y2		Y1(Cursor1) - Y2(Cursor2)		0.000		
		X Values	0.000	0.000	0.000		Y1 - Y1(Cursor1)	Y2 - Y2(Cursor2)	Max Y	Min Y	Avg Y
	CURSOR 1,2	Max(V(out))	1.0005p	1.0005p	0.000		0.000	0.000	1.0005p	1.0005p	1.0005p
		Max(I(RL))	0.9970	0.9970	0.000		0.9970	0.9970	0.9970	0.9970	0.9970

Figure 27. Output voltage and current values when the protection circuit is ON

As we can see from figure 26, the protection circuit works correctly because at high resistance values we have low current, and as long as the current is low, the voltage remains constant. When the resistance value decreases, the current starts to increase. It would increase to "infinity", and the voltage would still be constant, but the protection circuit prevents this. When the current increases above the limit value, the protection circuit comes into operation, automatically the output voltage starts to decrease, and the current will no longer increase, but will remain limited to the limit current of 1A. When the current is limited, the output voltage will tend to 0, as shown in figure 27.

Additionally, I ran only a Time Domain (Transient) analysis as in figure 24, manually setting the value of the parameter {R} of the load resistance to the value of 0Ω, thus short-circuiting the output and displaying the current through the load resistance to test and its limiting to 1A. And in this case the circuit works correctly, as we can see in figures 28 and 29.

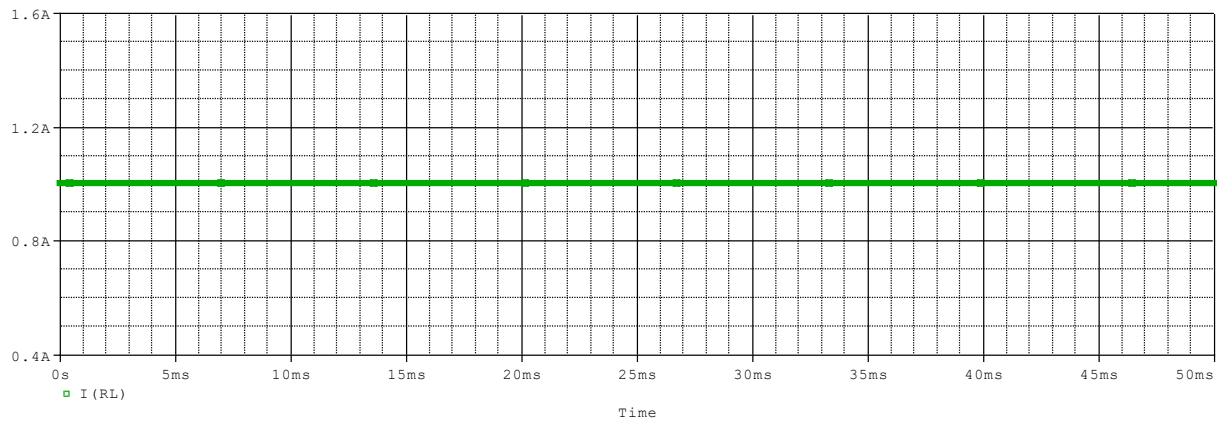


Figure 28. Current limitation at 1A when output is shorted

Probe Cursor

Trace Color	Trace Name	Y1	Y2	Y1 - Y2		Y1(Cursor1) - Y2(Cursor2)		-2.4340n		
	X Values	15.333m	0.000	15.333m		Y1 - Y1(Cursor1)	Y2 - Y2(Cursor2)	Max Y	Min Y	Avg Y
CURSOR 1,2	I(RL)	1.0005	1.0005	-2.4340n		0.000	0.000	1.0005	1.0005	1.0005

Figure 29. Current value when output is shorted

6.7. Worst Case analysis

I chose to perform a Worst Case analysis to see what happens with the circuit in the worst case in terms of tolerances, considering both LOT and device (DEV) tolerances. I will run this analysis for both the minimum output voltage and the maximum output voltage, with the resistor`s tolerance set on the circuit.

For this, we set a main Transient analysis with a simulation duration of 50ms (figure 30), then a secondary Worst Case analysis with a number of 5 runs (figure 31) because we have 4 components with tolerance and +1.

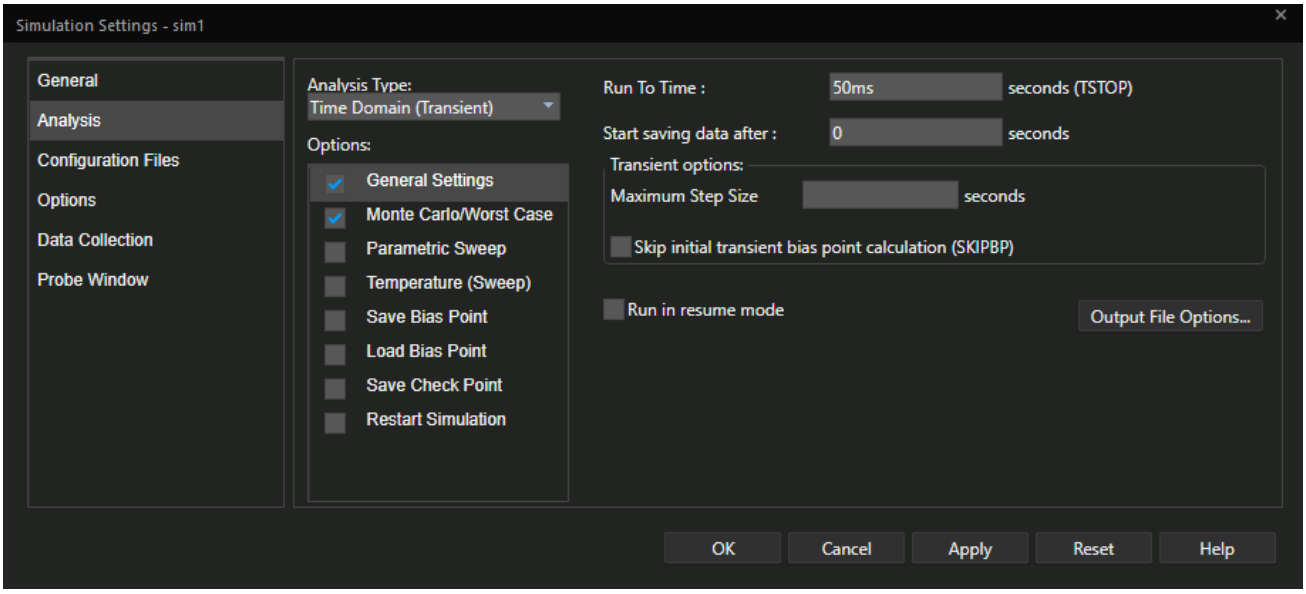


Figure 30. Transient analysis

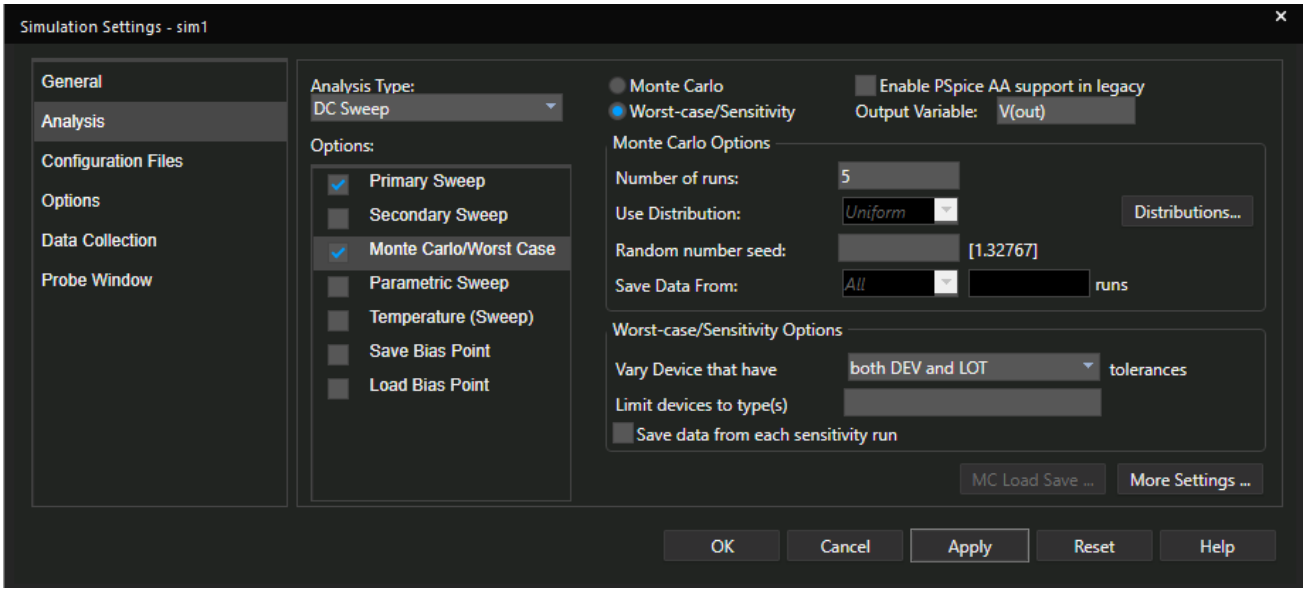


Figure 31. Worst Case analysis

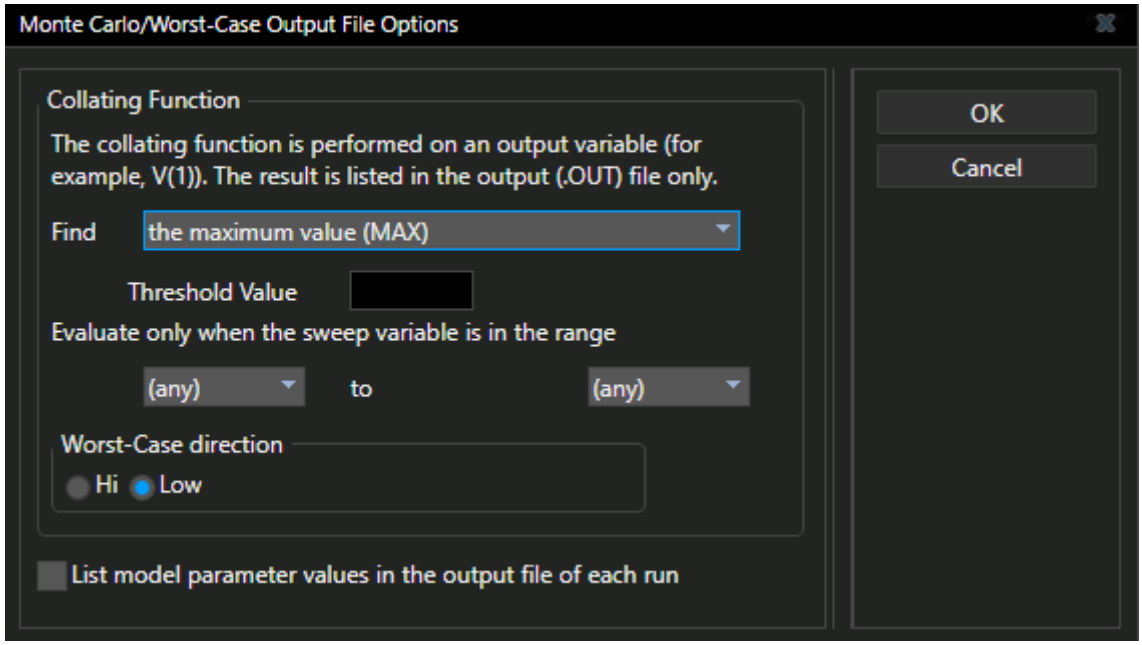


Figure 32. Setting the direction for the Worst Case analysis

First, we run this simulation for the minimum output voltage, once setting the Worst Case direction to Low (figure 32), meaning the resistors will have a value below the nominal value (tolerance -1%), then setting the direction to High (figure 32), when the resistors will have a value above the nominal value (tolerance +1%).

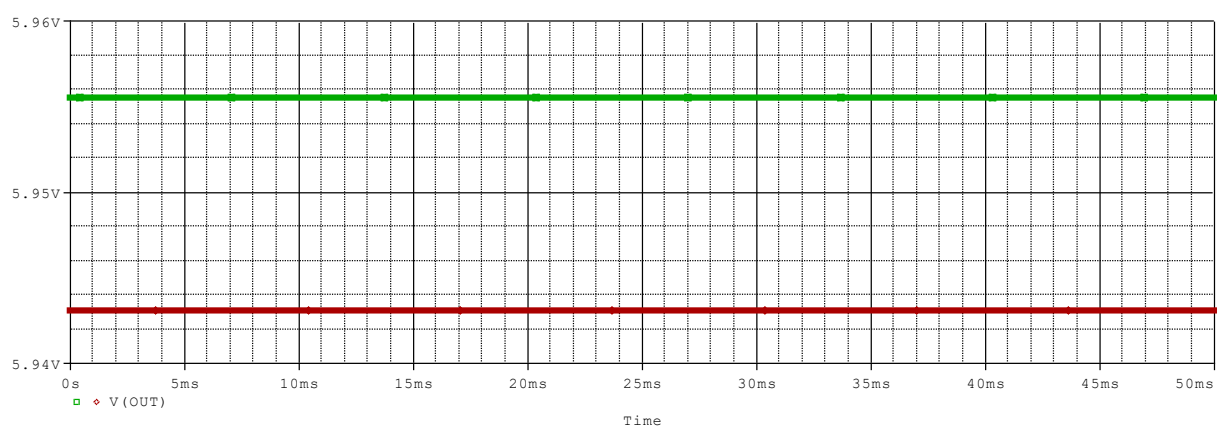


Figure 33. Worst Case analysis for V_{out_min} – Low

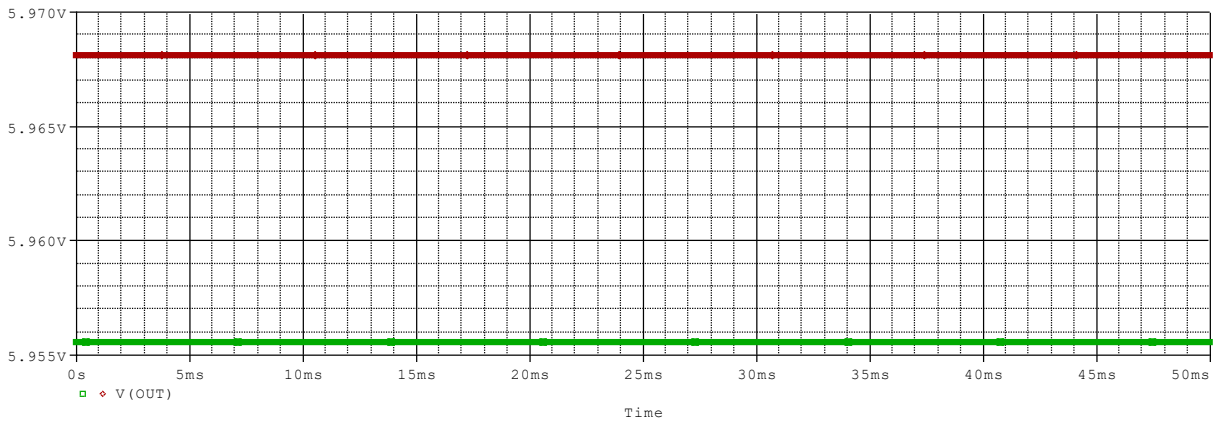


Figure 34. Worst Case analysis for V_{out_min} – High

For the Worst Case analysis of the maximum output voltage we followed the same steps as for the minimum voltage.

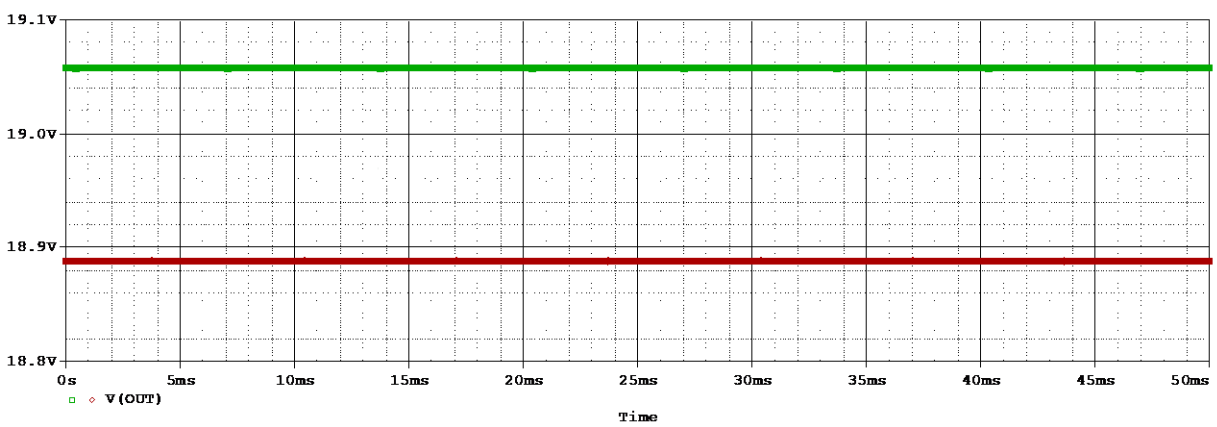


Figure 35. Worst Case analysis for V_{out_max} – Low

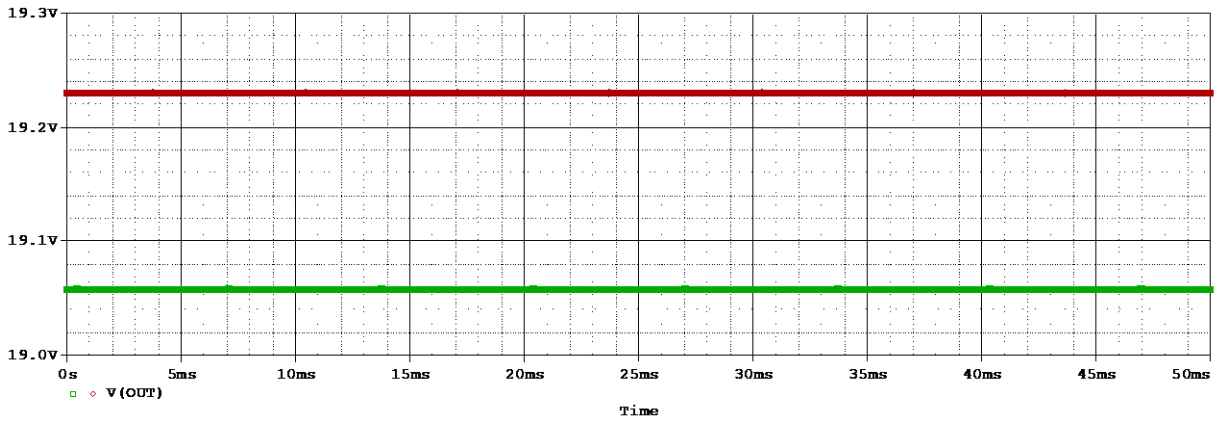


Figure 36. Worst Case analysis for V_{out_max} – High

As we can see from figures 33, 34, 35 and 36, the minimum and maximum output voltages move slightly up and down from the value they are supposed to stabilize. We can deduce that, in the worst case, the tolerances affect the proper functioning of the circuit.

6.8. Monte Carlo analysis

To analyze the circuit`s behavior taking into account the imperfections of the components, I decided to run a Monte Carlo analysis on both the minimum and the maximum output voltage. For this, from the simulation tab, we set the main analysis to Transient and in the secondary analysis tab we check the Monte Carlo option, where we choose a number of 60 runs (figure 37).

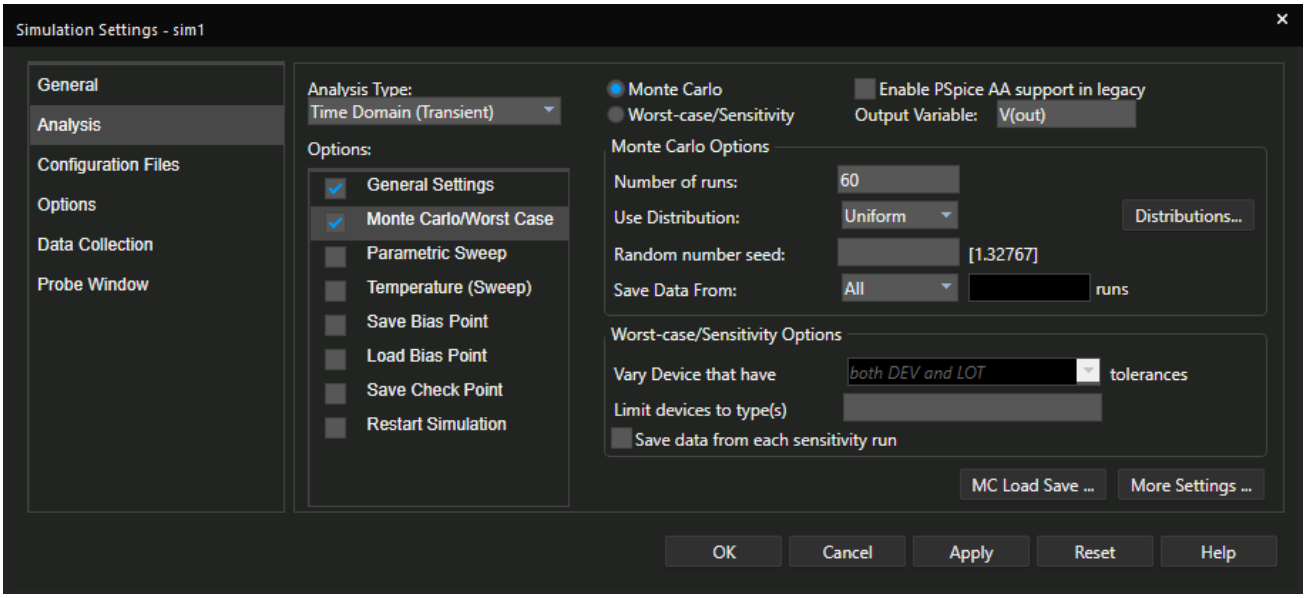


Figure 37. Monte Carlo analysis

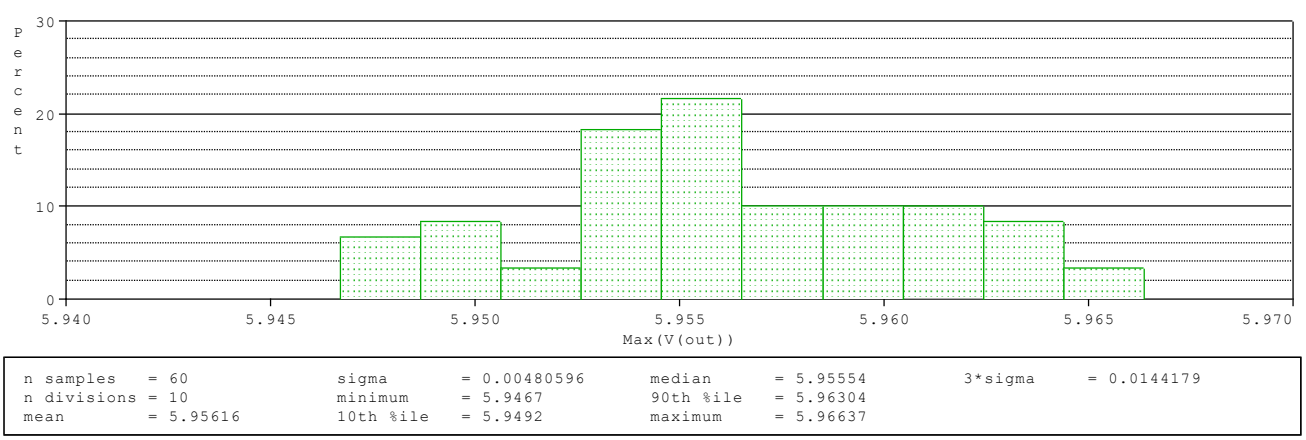


Figure 38. Histogram for the minimum output voltage

From figure 38 we can see that the minimum output voltage of the voltage regulator, in most cases, is in the range [5.953V; 5.954V], as well as in the range [5.954V; 5.9553V]. These two ranges would not affect much the correct operation of the circuit because the values are close to 6V.

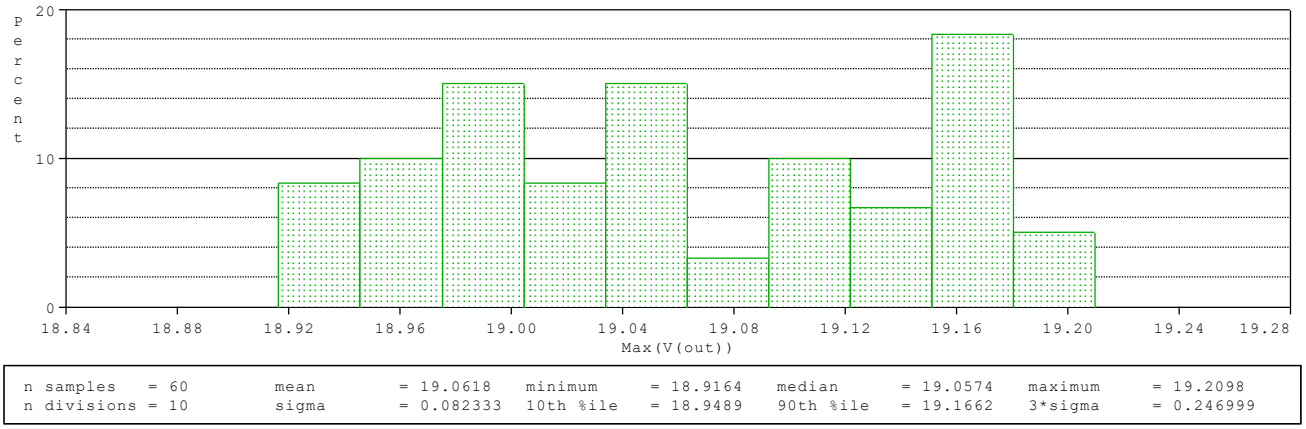


Figure 39. Histogram for the maximum output voltage

From figure 39 we observe that in most cases, the maximum output voltage is in the range [19.15V; 19.18V], which would lead to incorrect operation of the circuit because we are moving away from the value of 19V. On the other hand, there are considerable chances that the maximum output voltage will take values in the ranges [18.97V; 19.01V], respectively [19.03V; 19.06V], which would not affect the functionality of the circuit because we are very close to the maximum stabilization value.

6.9. Temperature analysis

I chose to also perform a temperature analysis since it was not mentioned whether the circuit would be used indoors or outdoors and I wanted to check if it would still function normally if exposed to different external factors. For this, I set a main Transient analysis with a simulation duration of 50ms, then a secondary Temperature Sweep analysis and ran the simulation at 3 different temperature values: 0degC, 15degC and 30degC (figure 40). I will run this analysis for both the minimum and maximum output voltage.

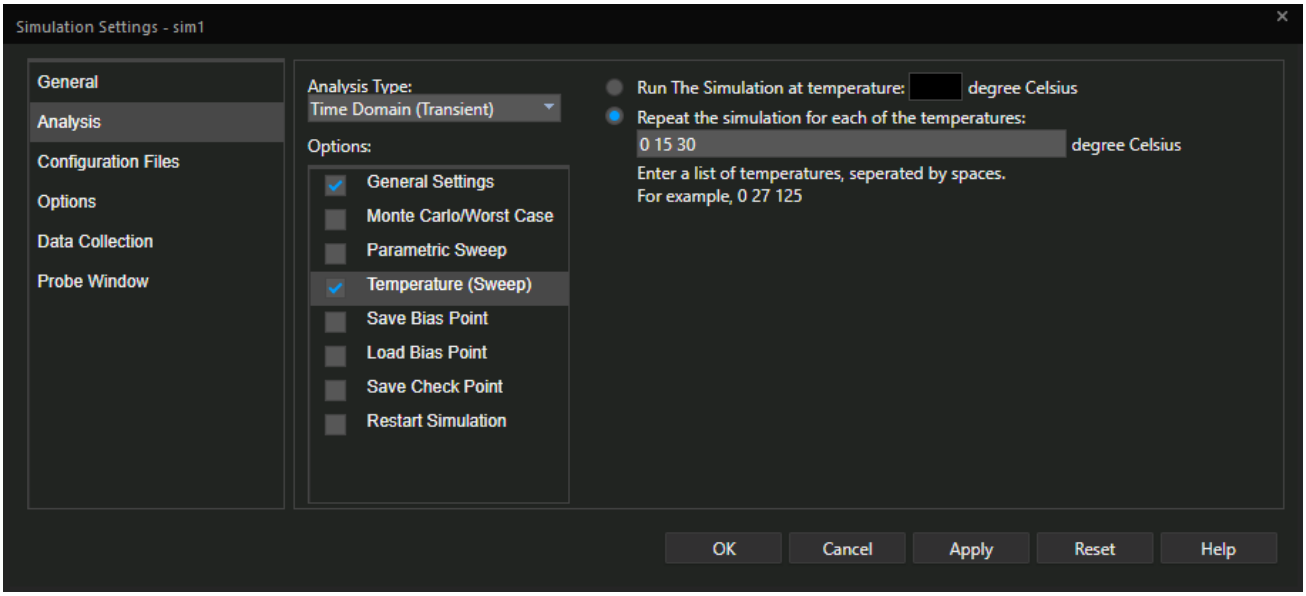


Figure 40. Temperature analysis

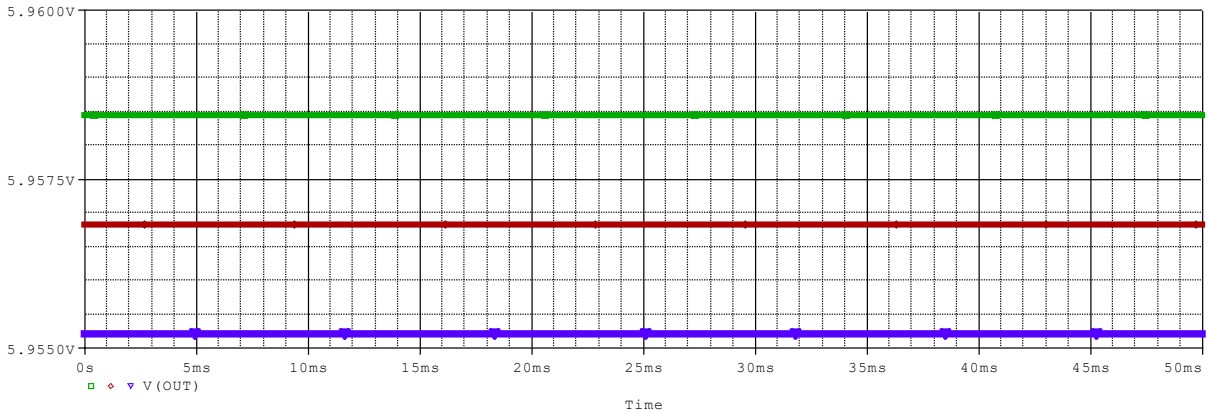


Figure 41. Minimum output voltage at 0°C (purple), 15°C (red), 30°C (green)

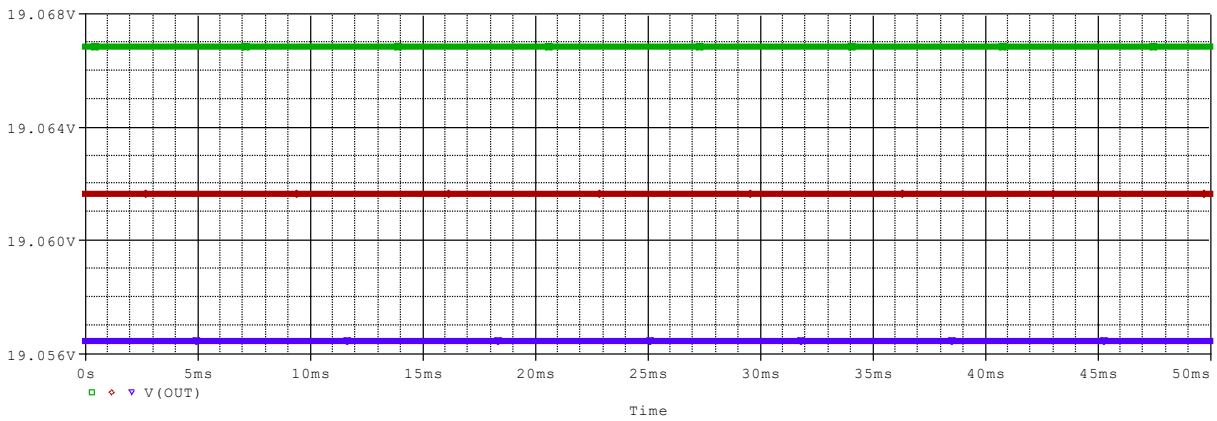


Figure 42. Maximum output voltage at 0°C (purple), 15°C (red), 30°C (green)

We can see from figures 41 and 42 that the output voltage varies very little with increasing temperature, so we can say that the circuit would work correctly regardless if its used indoors or outdoors.

7.0. Voltage Transfer characteristic

This characteristic helps us visualize how the output voltage varies with the input voltage. To generate this, we will set a main DC Sweep analysis on the input voltage, which we will sweep between 0V and 24V with a step of 0.1 (figure 43) and then plot V_{out}. We will run this simulation for both the minimum and the maximum output voltage.

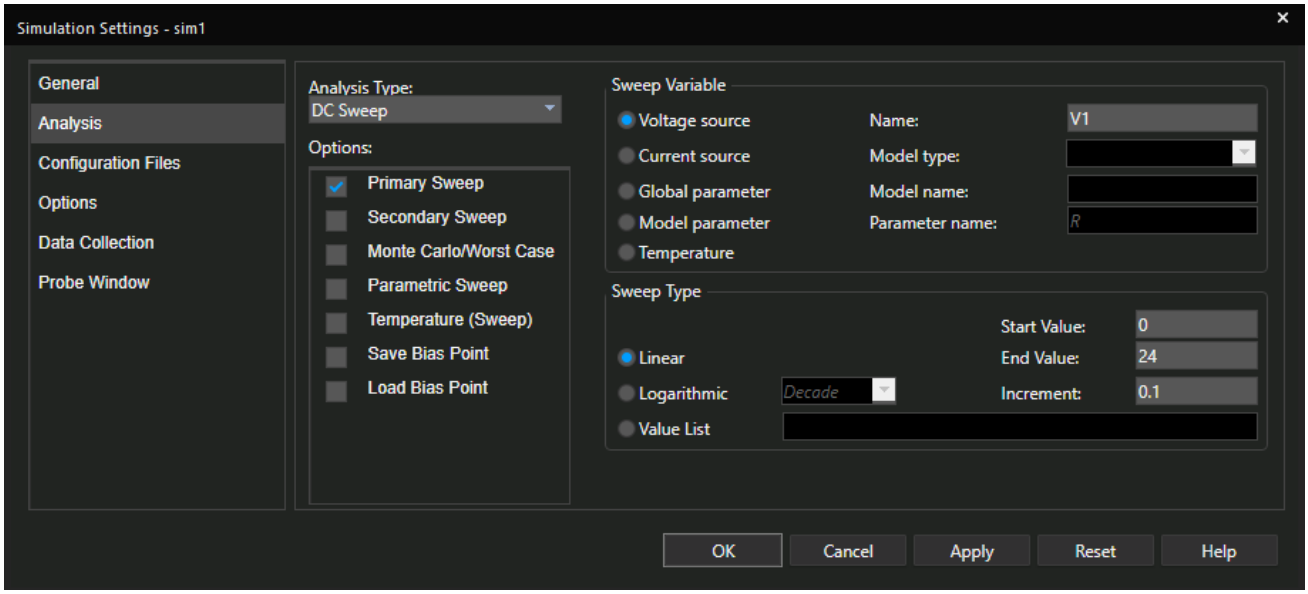


Figure 43. DC Sweep analysis

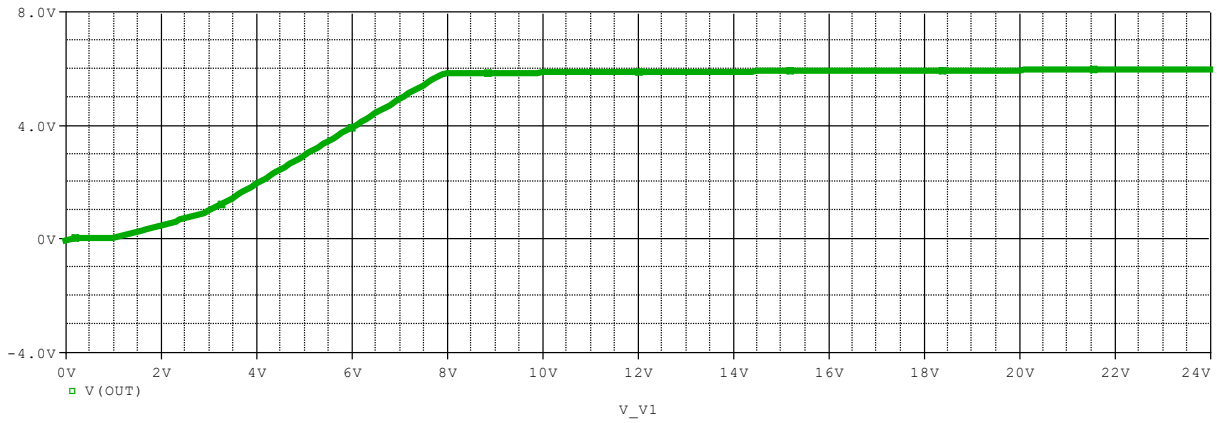


Figure 44. Voltage Transfer characteristic for V_{out_min}

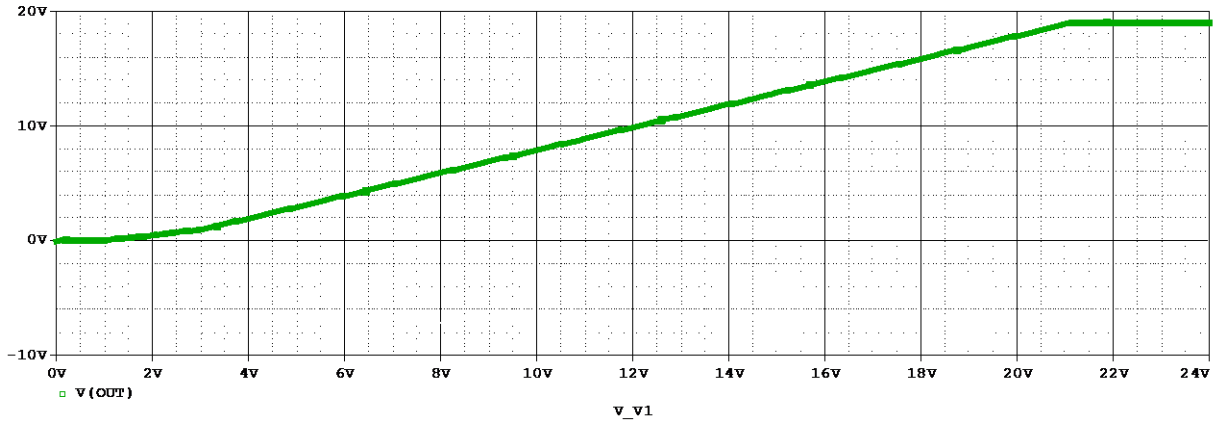


Figure 45. Voltage transfer characteristic for V_{out_max}

We can see from figures 44 and 45 that the output voltage, both minimum and maximum, increases with increasing input voltage, but at some point it is limited due to the stabilization circuit.

8.0. Design cost

As a final step, I decided to calculate the implementation cost of such a circuit. As a reference for the components, I used the Mouser Electronics website.

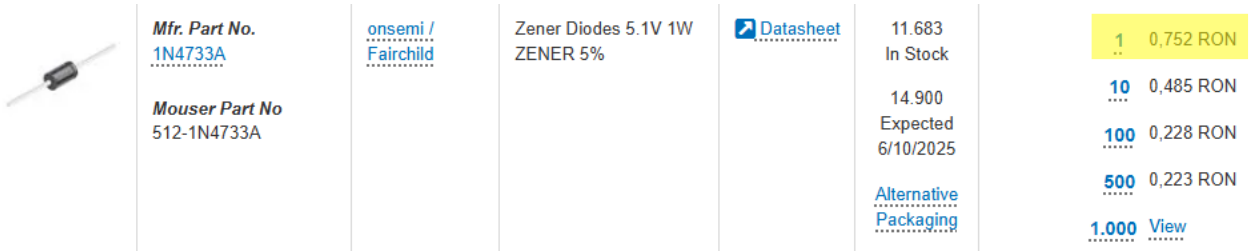
8.1. Input resistance

Mfr. Part No. LR1F383R *Mouser Part No* 279-LR1F383R	TE Connectivity / AMP	Metal Film Resistors - Through Hole LR1 1% 383R	Datasheet	7.556 In Stock	1 0,708 RON 10 0,584 RON 25 0,426 RON 100 0,233 RON Reel 4.000 0,094 RON 8.000 View

Figure 45. Input resistance cost

The price of such a resistor is 0.708 lei.

8.2. Zener diode



Mfr. Part No. 1N4733A *Mouser Part No* 512-1N4733A	onsemi / Fairchild	Zener Diodes 5.1V 1W ZENER 5%	Datasheet	11.683 In Stock 14.900 Expected 6/10/2025 Alternative Packaging	1 0,752 RON 10 0,485 RON 100 0,228 RON 500 0,223 RON 1.000 View

Figure 46. Zener diode cost

The price of this diode is 0.752 lei.

8.3. Error amplifier

+1 image	*Mfr. Part No.* TL082CPT *Mouser Part No* 511-TL082CP	STMicroelectronics	Operational Amplifiers - Op Amps Dual Gen Purp JFET Learn More	Datasheet	40.766 In Stock	Cut Tape 1 1,32 RON 10 1,01 RON 25 0,896 RON 100 0,837 RON Reel 4.000 0,777 RON 8.000 View

Figure 47. Error amplifier cost

Such an operational amplifier cost 1.32 lei.

8.4. Regulating transistor + protection transistor

	Mfr. Part No. 2N2222A *Mouser Part No* 494-2N2222A	Microchip Technology	Bipolar Transistors - BJT 40V 800mA 500mW Small-Signal BJT THT Learn More	Datasheet	3.876 In Stock 19.485 On Order View Dates	1 13,66 RON 100 12,67 RON
	Mfr. Part No. 2N3055G *Mouser Part No* 863-2N3055G	onsemi	Bipolar Transistors - BJT NPN 15A 60V	Datasheet	5.139 In Stock	1 23,51 RON 10 13,61 RON 50 12,72 RON 100 11,19 RON

Figure 48. Transistor`s cost

The price of these two transistors is 37.17 lei.

8.5. Protection resistance

	Mfr. Part No. RLC73K2ER649FTDF *Mouser Part No* 279-RLC73K2ER649FTDF	TE Connectivity / CGS	Current Sense Resistors - SMD RLC73K 2E R649 1% 1K RL Learn More	Datasheet	1.970 In Stock	Cut Tape 1 2,31 RON 10 1,75 RON 25 1,34 RON 100 0,757 RON

Figure 49. Protection resistance cost

The price of such a resistor is 2.31 lei.

8.6. Resistors from the feedback circuit

	Mfr. Part No. LR1F5K36 *Mouser Part No* 279-LR1F5K36	TE Connectivity / Holsworthy	Metal Film Resistors - Through Hole LR1 1% 5K36	Datasheet	6.234 In Stock	1 0,708 RON 10 0,584 RON 25 0,426 RON 100 0,233 RON
	Mfr. Part No. PT10MV10-223A1010-E-PM-S *Mouser Part No* 531-PT1MV1223A11EPMS	Amphenol Piher	Trimmer Resistors - Through Hole 10mm carbon through hole potentiometer Vertical adjust resistive value 22K Ohm linear taper tolerance 10% extended life	Datasheet	1.738 In Stock	1 3,69 RON 10 3,14 RON 25 2,94 RON 100 2,63 RON
	Mfr. Part No. PR01000101002FA100 *Mouser Part No* 594-PR01000101002FA1 New At Mouser	Vishay / BC Components	Metal Film Resistors - Through Hole PR01 1% A1 10K	Datasheet	819 In Stock	1 1,98 RON 10 1,45 RON 25 1,10 RON 100 0,648 RON

Figura 50. Resistors from the feedback circuit cost

These resistors have a total price of 6,378 lei.

In total, all the components in the circuit cost 48.64 lei.

9.0. BIBLIOGRAPHY

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[5] Standard resistor values table
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